

The Environmental Impact of Scalable Lung Cancer Classification Pipeline

How MLOps Helps Make AI Greener

MLOps is like the behind-the-scenes teamwork that brings together machine learning, DevOps (the stuff that keeps software running smoothly), and data engineering to make AI projects easier to build, launch, and keep an eye on. It can also help make AI way more eco-friendly by cutting down on energy use and carbon emissions. In below it is explained in very simple word:

1. Using Just What We Need (No Waste!)

MLOps uses tools like Kubernetes to smartly manage computing power, so we only use what we need for our AI projects. This can cut down energy waste by up to 40%, in addition with tools like Terraform, we can set up cloud systems to automatically shut off when they're not in use and even pick locations that use cleaner energy, saving both money and the planet[1].

2. Automating to Save Time and Energy

Think of MLOps like a super-organized friend who plans your day so you don't waste time or gas driving around aimlessly. It uses automated systems (like GitHub Actions) to speed up the process of getting AI models ready to go. But here's the green twist—it can also schedule these tasks for times when the power grid is using more renewable energy, like when the sun's shining for solar power. This not only gets things done faster but also lowers the carbon footprint of our AI work—like choosing to carpool instead of driving alone[1].

3. Keeping an Eye on Our Carbon Footprint

MLOps is like having a fitness tracker for our AI project, but instead of counting steps, it tracks how much energy and CO2 our project is using. Tools like Prometheus, Grafana, or AWS CloudWatch can show us exactly where we're using too much power. By fixing these inefficiencies (like shutting down extra servers we don't need), we can save a ton of energy and reduce our environmental impact, all while keeping costs down[1][2].

4. Making AI Models Lighter and Greener

Imagine if we could pack for a trip with a smaller suitcase that still fits everything you need. MLOps helps do that for AI models by using tricks like model pruning (trimming away unnecessary parts), quantization (simplifying the math), and federated learning (training models locally so we don't have to send tons of data over the internet). These lighter models use less power to run, which means fewer emissions—like switching from a big gas-guzzling truck to a zippy electric scooter[3].

5. Being Responsible and Fair

MLOps isn't just about saving energy—it's also about doing AI the right way. It helps us build AI that's fair, doesn't have biases, and thinks about the bigger picture, like sustainability and ethics[3].

MLOps Impact of our ML Application

This part looks at the environmental impact of our machine learning project called the Scalable Lung Cancer Classification pipeline, which helps identify types of lung cancer using gene data. It uses tools like Dask and XGBoost to process data efficiently, but we wanted to see how much energy it uses and how much carbon dioxide (CO₂) it releases—kind of like checking the "carbon footprint" of a road trip.

What We Found

- **Energy Use and Carbon Emissions:** The project uses a small amount of energy—about 0.0759 kWh (like running a light bulb for a few hours) for 10 runs, releasing 28.4 grams of CO₂. That's tiny, like driving a car just 0.016 km or a super short flight (0.003% of a Paris-Dublin trip) [7].

- **Why It's Low:** The project uses a small dataset and runs on a regular laptop (Intel Core i5, no fancy GPU), so it doesn't need much power. It only takes about 193 seconds to finish one run.

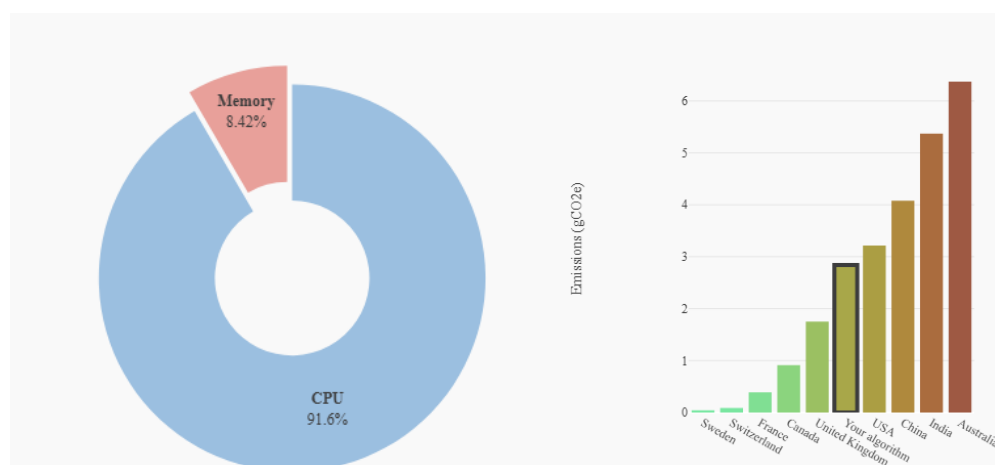
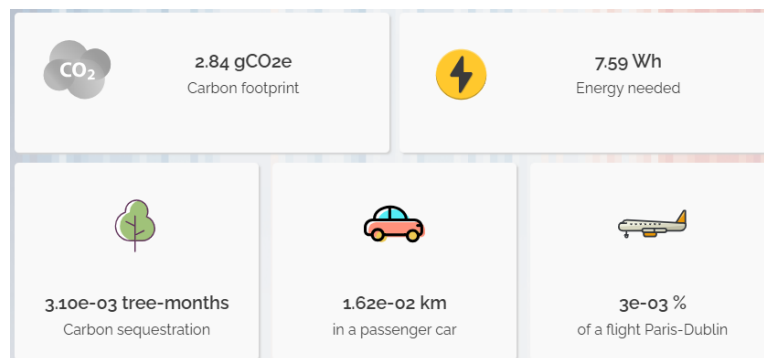
- **But There's a Catch:** Even small projects like this can add up if we run them a lot or use bigger datasets. For example, in other fields like language AI, one model can release over 750,000

grams of CO₂—yikes! [1] The tech industry as a whole contributes 2% of global CO₂ emissions, and that's growing as more projects use machine learning [5].

How We Can Make It Greener

We came up with some practical ideas to lower the environmental impact, inspired by experts in sustainable computing and MLOps (a way to manage machine learning projects). Here's what we can do:

1. Work Smarter with Dask: We can tweak how Dask splits up the work to match our laptop's 12 cores better, so it runs faster and uses less energy. This fits into the "Data Acquisition" and "Experimentation" steps of a good MLOps plan [2, 3]. It could save 10-20% of energy.



2. Simplify the Model: We can cut down the number of genes we analyze (from 43,488 to fewer) and make the XGBoost model less complex. This is part of the "ML Training" step in MLOps and could save 20-30% of energy [2, 3].

3. Use Less Power-Hungry Math: By using simpler math (like float16 instead of float32), we can lower the energy our laptop needs—kind of like switching to a fuel-efficient car. This could save 10-15% of energy [7].

4. Pick Greener Tech: If we scale up, we can use energy-efficient cloud servers (like ARM-based ones) or run the project in places with cleaner energy (like where wind or solar power is used). This matches the "Sustainability" step in MLOps and could cut emissions by up to 50% [1, 3].

5. Be Efficient with Runs: We can use tools like MLflow to avoid repeating experiments and stop training early if the model isn't improving. This fits the "Continuous Training" and "Monitoring" steps in MLOps and could reduce runs by 20-30%, saving more energy [5][6].

6. Keep an Eye on Emissions: We can add a tool called CodeCarbon to track how much CO2 we're releasing while the project runs. It's like putting a "fuel gauge" on our project to stay mindful, and it's part of the "Sustainability" focus in MLOps [5][6].

Why This Matters

Even though our project's carbon footprint is small, every bit counts—especially in healthcare, where we want to help patients without harming the planet. By following these steps, we can cut emissions by 30-50%, making our project not just smart but also kind to the environment. It's like choosing to walk instead of drive for short trips—it all adds up!

References

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